



# ITSimplera Institute

## **WEEK 02: FLSM, VLSM, and VLAN Configuration on Packet Tracer and GNS3**

**Domain:** Network administration

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## **Introduction:**

This report offers a practical study of IP subnetting techniques and VLAN implementation, demonstrating how logical network segmentation can be applied to real-world small scale scenarios. In the first part of the report, Fixed Length Subnet Masking (FLSM) is discussed where we split one Class C network into subnets of equal size, with examples of a small office, a school, a clinic and a cafe chain. The report then proceeds to Variable-Length Subnet Masking (VLSM) which is a more efficient technique that assigns subnet sizes according to the actual host requirements of each department or location, thus reducing address wastage. It is shown in five examples, such as a small business, clinic, school, retail store, and apartment building, with departments having different numbers of hosts. The mask is calculated for all subnetting scenarios at the binary level. Resulting subnet masks are shown in CIDR and dotted-decimal notation. Full IP allocation tables are shown with network addresses, usable addresses, and broadcast addresses.

Besides subnetting, this report also explains the concept of VLAN which creates isolated broadcast domains over a shared switch infrastructure. This is implemented by using two simulated network topologies, one in Cisco Packet Tracer and another in GNS3, each having two switches connected through a trunk link with VLAN 10 and VLAN 20 configured across multiple PCs.

## **Fixed length Subnet Mask:**

A Fixed-Length Subnet Mask (FLSM) is an IP addressing technique where a large network block is divided into multiple smaller subnets, all of which use the identical subnet mask and hold the exact same number of IP addresses.

## **FLSM scenarios:**

### **1) Small office network divided into 4 equal subnets:**

The office needs to use IP 192.168.5.0/24 and convert it into 4 equal subnets.

### **Subnet mask calculation:**

**192.168.5.0/24 → 4 subnets**

- Subnets Required = 4 →  $2^2 = 4$ , borrow 2 bits

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11000000 (/26)
- Subnet mask = **255.255.255.192**

Hosts per subnet  $2^6 - 2 = 62$

IP allocation:

| Subnets  | Network address  | Useable Address Range | Broadcast Address |
|----------|------------------|-----------------------|-------------------|
| <b>1</b> | 192.168.5.0/26   | 192.168.5.1-62        | 192.168.5.63      |
| <b>2</b> | 192.168.5.64/26  | 192.168.5.65-126      | 192.168.5.127     |
| <b>3</b> | 192.168.5.128/26 | 192.168.5.129-190     | 192.168.5.191     |
| <b>4</b> | 192.168.5.192/26 | 192.168.5.193-254     | 192.168.5.255     |

## 2) Small school with 3 subnets each with 30 hosts:

There is a small school with 3 classrooms. Each classroom has 30 users

**Subnet mask calculation:**

**192.168.20.0/24 → 3 subnets (round up to 4)**

- Subnets Required  $\geq 3 \rightarrow 2^2 = 4$ , borrow 2 bits
- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11000000 (/26)
- Subnet mask = **255.255.255.192**

Hosts per subnet  $2^6 - 2 = 62$  (only 30 Required there is room for growth)

IP allocation:

| Subnets       | Network address  | Useable Address Range | Broadcast address |
|---------------|------------------|-----------------------|-------------------|
| <b>Room A</b> | 192.168.5.0/26   | 192.168.5.1-62        | 192.168.5.63      |
| <b>Room B</b> | 192.168.5.64/26  | 192.168.5.65-126      | 192.168.5.127     |
| <b>Room C</b> | 192.168.5.128/26 | 192.168.5.129-190     | 192.168.5.191     |

**3) Small clinic network divided into 6 subnets each with 25 hosts:**

There are six clinics in total each clinic with 30 users

**Subnet mask calculation:**

**192.168.30.0/24 → 6 subnets (round up to 8)**

- Subnets Required  $\geq 6 \rightarrow 2^3 = 8$ , borrow 3 bits
- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**111**00000 (/27)
- Subnet mask = **255.255.255.224**

Hosts per subnet  $2^5 - 2 = 30$  (only 25 are Required) IP allocation:

| Subnets  | Network address   | Useable address Range | Broadcast address |
|----------|-------------------|-----------------------|-------------------|
| <b>1</b> | 192.168.30.0/27   | 192.168.30.1-30       | 192.168.30.31     |
| <b>2</b> | 192.168.30.32/27  | 192.168.30.33-62      | 192.168.30.63     |
| <b>3</b> | 192.168.30.64/27  | 192.168.30.64-94      | 192.168.30.95     |
| <b>4</b> | 192.168.30.96/27  | 192.168.30.97-126     | 192.168.30.127    |
| <b>5</b> | 192.168.30.128/27 | 192.168.30.129-158    | 192.168.30.159    |
| <b>6</b> | 192.168.30.160/27 | 192.168.30.161-191    | 192.168.30.192    |

**4) Small cafe chain divided into 8 equal subnets:**

There is a small cafe chain consisting of 8 cafes, each cafe requires 14 hosts including the 2 links

**Subnet mask calculation:**

192.168.40.0/24 → 8 subnets

- Subnets Required = 8 →  $2^3 = 8$ , borrow 3 bits
- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**111**00000 (/27)
- Subnet mask = **255.255.255.240**

Hosts/subnet:  $2^4 - 2 = 14$

IP allocation:

| Subnet | Network address   | Usable address range | Broadcast address |
|--------|-------------------|----------------------|-------------------|
| 1      | 192.168.40.0/28   | 192.168.40.1–14      | 192.168.40.15     |
| 2      | 192.168.40.16/28  | 192.168.40.17–30     | 192.168.40.31     |
| 3      | 192.168.40.32/28  | 192.168.40.33–46     | 192.168.40.47     |
| 4      | 192.168.40.48/28  | 192.168.40.49–62     | 192.168.40.63     |
| 5      | 192.168.40.64/28  | 192.168.40.65–78     | 192.168.40.79     |
| 6      | 192.168.40.80/28  | 192.168.40.81–94     | 192.168.40.95     |
| 7      | 192.168.40.96/28  | 192.168.40.97–110    | 192.168.40.111    |
| 8      | 192.168.40.112/28 | 192.168.40.113–126   | 192.168.40.127    |

### Variable Length Subnet Mask:

Variable Length Subnet Mask (VLSM) is a subnetting technique that allows network administrators to divide an IP network into subnets of different sizes. Instead of using one fixed subnet mask for all subnets (as in FLSM), VLSM allows you to apply multiple subnet masks within the same network, based on the number of hosts each subnet requires.

### CIDR reference table:

| CIDR | Binary (last octet) | Decimal Mask    |
|------|---------------------|-----------------|
| /24  | 00000000            | 255.255.255.0   |
| /25  | 10000000            | 255.255.255.128 |
| /26  | 11000000            | 255.255.255.192 |
| /27  | 11100000            | 255.255.255.224 |
| /28  | 11110000            | 255.255.255.240 |
| /29  | 11111000            | 255.255.255.248 |
| /30  | 11111100            | 255.255.255.252 |

**VLSM scenarios:****1) A small business network:**

Host requirements: Sales=40, IT=20, HR=8, WAN Link=2

IP address: 196.168.1.0/24

**Subnet mask calculation:**

**Sales:** Hosts Required = 40.  $2^6 - 2 = 62 \geq 40$  (borrow 2 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11000000 (/26)
- Subnet mask: **255.255.255.192**

**IT:** Hosts Required = 20.  $2^5 - 2 = 30 \geq 20$  (borrow 3 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11100000 (/27)
- Subnet mask: **255.255.255.224**

**Human Resources:** Hosts Required = 8,  $2^4 - 2 = 14 \geq 8$  (borrow 4 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11110000 (/28)
- Subnet mask: **255.255.255.240**

**WAN-LINK:** Hosts Required = 2.  $2^2 - 2 = 2 \geq 2$  (borrow 6 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11111100 (/30)
- Subnet mask: **255.255.255.252**

IP allocation:

| Subnet | Network Address  | Usable address range | Broadcast address | Subnet Mask            |
|--------|------------------|----------------------|-------------------|------------------------|
| Sales  | 192.168.1.0/26   | 192.168.1.1–62       | 192.168.1.63      | <b>255.255.255.192</b> |
| IT     | 192.168.1.64/27  | 192.168.1.65–94      | 192.168.1.95      | <b>255.255.255.224</b> |
| HR     | 192.168.1.96/28  | 192.168.1.97–110     | 192.168.1.111     | <b>255.255.255.240</b> |
| WAN    | 192.168.1.112/30 | 192.168.1.113.114    | 192.168.1.115     | <b>255.255.255.252</b> |

## 2) Small Clinic Network:

Host Requirements: Reception=30, Doctors=15, Pharmacy=10, Server Link=2

IP address: 192.168.2.0/24

### Subnet mask calculation:

**Reception:** Hosts required=30.  $2^5 - 2 = 30 \geq 30$  (borrow 3 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**111**00000 (/27)
- Subnet mask: **255.255.255.224**

**Doctors:** Hosts required=15.  $2^5 - 2 = 30 \geq 15$  (borrow 3 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**111**00000 (/27)
- Subnet mask: **255.255.255.224**

**Pharmacy:** Hosts Required = 10,  $2^4 - 2 = 14 \geq 10$  (borrow 4 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**1111**0000 (/28)
- Subnet mask: **255.255.255.240**

**Server-LINK:** Hosts Required = 2.  $2^2 - 2 = 2 \geq 2$  (borrow 6 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)



- New mask: 11111111.11111111.11111111.**11111100** (/30)
- Subnet mask: **255.255.255.252**

IP allocation:

| Subnet             | Network Address | Useable Address Range | Broadcast Address | Subnet mask            |
|--------------------|-----------------|-----------------------|-------------------|------------------------|
| <b>Reception</b>   | 192.168.2.0/27  | 192.168.2.1–.30       | 192.168.2.31      | <b>255.255.255.224</b> |
| <b>Doctors</b>     | 192.168.2.32/27 | 192.168.2.33–.62      | 192.168.2.63      | <b>255.255.255.224</b> |
| <b>Pharmacy</b>    | 192.168.2.64/28 | 192.168.2.65–.78      | 192.168.2.79      | <b>255.255.255.240</b> |
| <b>Server Link</b> | 192.168.2.80/30 | 192.168.2.81–.82      | 192.168.2.83      | <b>255.255.255.252</b> |

### 3) Small School Network:

Hosts Requirements: Classrooms=50, Staff Room=25, Library=12, Admin=6, Router Link=2

Ip address: 192.168.3.0/24

#### Subnet mask calculation:

**Classrooms:** Hosts Required = 50.  $2^6 - 2 = 62 \geq 50$  (borrow 2 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**11000000** (/26)
- Subnet mask: **255.255.255.192**

**Staff Room:** Hosts required=25.  $2^5 - 2 = 30 \geq 25$  (borrow 3 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**11100000** (/27)
- Subnet mask: **255.255.255.224**

**Library:** Hosts Required = 12.  $2^4 - 2 = 14 \geq 12$  (borrow 4 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**11110000** (/28)
- Subnet mask: **255.255.255.240**

**Admin:** Hosts Required = 6.  $2^3 - 2 = 6 \geq 6$  (borrow 5 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**11111000** (/29)
- Subnet mask: **255.255.255.248**

**Router-link:** Hosts Required = 2.  $2^2 - 2 = 4 \geq 2$  (borrow 6 bits from default mask)

- Default mask: 11111111.11111111.11111111.**11111100** (/24)
- New mask: 11111111.11111111.11111111.**11111100** (/30)
- Subnet mask: **255.255.255.252**

IP allocation:

| Subnet             | Network address  | Useable address Range | Broadcast address | Subnet mask            |
|--------------------|------------------|-----------------------|-------------------|------------------------|
| <b>Classrooms</b>  | 192.168.3.0/26   | 192.168.3.1–.62       | 192.168.3.63      | <b>255.255.255.192</b> |
| <b>Staff Room</b>  | 192.168.3.64/27  | 192.168.3.65–.94      | 192.168.3.95      | <b>255.255.255.224</b> |
| <b>Library</b>     | 192.168.3.96/28  | 192.168.3.97–.110     | 192.168.3.111     | <b>255.255.255.240</b> |
| <b>Admin</b>       | 192.168.3.112/29 | 192.168.3.113–.118    | 192.168.3.119     | <b>255.255.255.248</b> |
| <b>Router Link</b> | 192.168.3.120/30 | 192.168.3.121–.122    | 192.168.3.123     | <b>255.255.255.252</b> |

#### 4) Small Retail Shop Network:

Host Requirements: Sales Floor=35, Stockroom=15, Manager Office=5, POS Link=2

IP address: 192.168.4.0/24

**-Subnet mask calculation:**

**Sales Floor:** Hosts Required = 35.  $2^6 - 2 = 62 \geq 35$  (borrow 2 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**11000000** (/26)
- Subnet mask: **255.255.255.192**

**Stock Room:** Hosts required=15.  $2^5 - 2 = 30 \geq 15$  (borrow 3 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**11100000** (/27)
- Subnet mask: **255.255.255.224**

**Manager office:** Hosts Required = 5.  $2^3 - 2 = 6 \geq 5$  (borrow 5 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.**11111000** (/29)
- Subnet mask: **255.255.255.248**

**Pos-link:** Hosts Required = 2.  $2^2 - 2 = 4 \geq 2$  (borrow 6 bits from default mask)

- Default mask: 11111111.11111111.11111111.**11111100** (/24)
- New mask: 11111111.11111111.11111111.**11111100** (/30)
- Subnet mask: **255.255.255.252**

Ip allocation:

| Subnet                | Network address  | Useable address Range | Broadcast Address | Subent Mask            |
|-----------------------|------------------|-----------------------|-------------------|------------------------|
| <b>Sales Floor</b>    | 192.168.4.0/26   | 192.168.4.1–.62       | 192.168.4.63      | <b>255.255.255.192</b> |
| <b>Stockroom</b>      | 192.168.4.64/27  | 192.168.4.65–.94      | 192.168.4.95      | <b>255.255.255.224</b> |
| <b>Manager Office</b> | 192.168.4.96/29  | 192.168.4.97–.102     | 192.168.4.103     | <b>255.255.255.248</b> |
| <b>POS Link</b>       | 192.168.4.104/30 | 192.168.4.105.106     | 192.168.4.107     | <b>255.255.255.252</b> |

### 5) Small Apartment Complex Network:

Requirements: Building A=45, Building B=28, Building C=14, Management Office=6, ISP Link=2

IP addressing: 192.168.5.0/24

**Subnet Mask calculation:**

**Building A:** Hosts Required = 45.  $2^6 - 2 = 62 \geq 45$  (borrow 2 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11000000 (/26)
- Subnet mask: **255.255.255.192**

**Building B:** Hosts required=28.  $2^5 - 2 = 30 \geq 28$  (borrow 3 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11100000 (/27)
- Subnet mask: **255.255.255.224**

**Building C:** Hosts Required = 14.  $2^4 - 2 = 14 \geq 14$  (borrow 4 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11110000 (/28)
- Subnet mask: **255.255.255.240**

**Management office:** Hosts Required = 6.  $2^3 - 2 = 6 \geq 6$  (borrow 5 bits from default mask)

- Default mask: 11111111.11111111.11111111.00000000 (/24)
- New mask: 11111111.11111111.11111111.11111000 (/29)
- Subnet mask: **255.255.255.248**

**Isp-link:** Hosts Required = 2.  $2^2 - 2 = 4 \geq 2$  (borrow 6 bits from default mask)

- Default mask: 11111111.11111111.11111111.11111100 (/24)
- New mask: 11111111.11111111.11111111.11111100 (/30)
- Subnet mask: **255.255.255.252**

#### IP Allocation:

| Subnets           | Network address | Useable address Range | Broadcast address | Subnet mask            |
|-------------------|-----------------|-----------------------|-------------------|------------------------|
| <b>Building A</b> | 192.168.5.0/26  | 192.168.5.1–62        | 192.168.5.63      | <b>255.255.255.192</b> |
| <b>Building B</b> | 192.168.5.64/27 | 192.168.5.65–<br>.94  | 192.168.5.95      | <b>255.255.255.192</b> |

|                          |                  |                        |               |                        |
|--------------------------|------------------|------------------------|---------------|------------------------|
| <b>Building C</b>        | 192.168.5.96/28  | 192.168.5.97–<br>.110  | 192.168.5.111 | <b>255.255.255.192</b> |
| <b>Management Office</b> | 192.168.5.112/29 | 192.168.5.113–<br>.118 | 192.168.5.119 | <b>255.255.255.192</b> |
| <b>ISP Link</b>          | 192.168.5.120/30 | 192.168.5.121–<br>.122 | 192.168.5.123 | <b>255.255.255.192</b> |

### **VLAN (virtual local area network):**

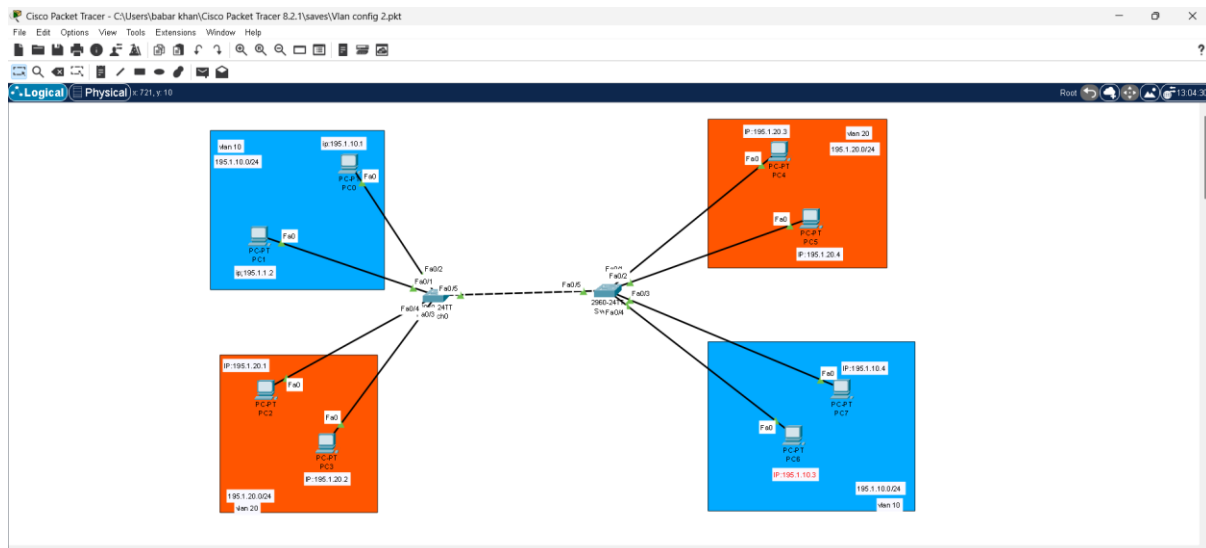
A VLAN (Virtual Local Area Network) is a logical grouping of devices on a switch that behaves as if it were its own separate physical network, even though the devices may be connected to the same physical switch or set of switches. Normally, all ports on a switch belong to a single broadcast domain, meaning every device can potentially see every other device's broadcast traffic. VLANs divide a switch (or group of switches) into multiple separate broadcast domains at Layer 2, so that traffic from one VLAN is isolated from another.

### **Key characteristics of VLANs:**

- Each VLAN is assigned a unique VLAN ID (e.g., VLAN 10, VLAN 20).
- Ports on a switch are assigned to a specific VLAN (access ports) or carry multiple VLANs between switches (trunk ports).
- Devices in the same VLAN can communicate directly, as if on the same physical switch.
- Devices in different VLANs cannot communicate with each other unless a Layer 3 device (a router, or a Layer 3 switch performing "router-on-a-stick" or inter-VLAN routing) is configured to pass traffic between them.
- VLANs improve network security, reduce broadcast traffic, and allow logical segmentation (e.g., separating guest traffic from staff traffic) without needing separate physical switches.

## Network Topology Cisco packet tracer:

The topology consists of two switches (Switch0 and Switch1) connected via a trunk link with four PCs attached to each switch. Two VLANs were configured: VLAN 10 and VLAN 20, each spanning both switches.



## IP allocation of different VLANs:

The table below summarizes the VLAN membership and IP addressing scheme observed in the topology diagram. VLAN 10 uses the 195.1.10.0/24 subnet and VLAN 20 uses the 195.1.20.0/24 subnet.

| VLAN ID | VLAN Name | Subnet        | PC  | Assigned IP |
|---------|-----------|---------------|-----|-------------|
| 10      | guest1    | 195.1.10.0/24 | PC0 | 195.1.10.1  |
| 10      | guest1    | 195.1.10.0/24 | PC1 | 195.1.1.2 * |
| 10      | guest1    | 195.1.10.0/24 | PC6 | 195.1.10.3  |
| 10      | guest1    | 195.1.10.0/24 | PC7 | 195.1.10.4  |
| 20      | guest2    | 195.1.20.0/24 | PC2 | 195.1.20.1  |
| 20      | guest2    | 195.1.20.0/24 | PC3 | 195.1.20.2  |
| 20      | guest2    | 195.1.20.0/24 | PC4 | 195.1.20.3  |
| 20      | guest2    | 195.1.20.0/24 | PC5 | 195.1.20.4  |

## Showing VLANs on both switches:

Both switches were checked using the "show vlan" command to confirm VLAN creation and port assignment.

### Switch0 — VLAN table:

- VLAN 10 (guest1): Fa0/1, Fa0/2
- VLAN 20 (guest2): Fa0/3, Fa0/4

### Switch1 — VLAN table:

- VLAN 10 (guest1): Fa0/1, Fa0/2
- VLAN 20 (guest2): Fa0/3, Fa0/4

```
Switch>
Switch>enable
Switch#config terminal
Switch(config)#
Switch(config)#
Switch(config)#do show vlan
Switch(config)#do show vlan
```

| VLAN Name               | Status | Ports                                                                                                                                                                        |
|-------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/6, Fa0/7, Fa0/8, Fa0/9<br>Fa0/10, Fa0/11, Fa0/12, Fa0/13<br>Fa0/14, Fa0/15, Fa0/16, Fa0/17<br>Fa0/18, Fa0/19, Fa0/20, Fa0/21<br>Fa0/22, Fa0/23, Fa0/24, Gig0/1<br>Gig0/2 |
| 10 guest1               | active | Fa0/1, Fa0/2                                                                                                                                                                 |
| 20 guest2               | active | Fa0/3, Fa0/4                                                                                                                                                                 |
| 1002 fddi-default       | active |                                                                                                                                                                              |
| 1003 token-ring-default | active |                                                                                                                                                                              |
| 1004 fddinet-default    | active |                                                                                                                                                                              |
| 1005 trnet-default      | active |                                                                                                                                                                              |

| VLAN Type | SAID | MTU    | Parent | RingNo | BridgeNo | Stp | BrdgMode | Trans1 | Trans2 |
|-----------|------|--------|--------|--------|----------|-----|----------|--------|--------|
| 1         | enet | 100001 | 1500   | -      | -        | -   | -        | 0      | 0      |
| 10        | enet | 100010 | 1500   | -      | -        | -   | -        | 0      | 0      |
| 20        | enet | 100020 | 1500   | -      | -        | -   | -        | 0      | 0      |
| 1002      | fddi | 101002 | 1500   | -      | -        | -   | -        | 0      | 0      |

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## Ping tests Results:

Ping tests were run from a PC on VLAN 10 to verify connectivity within and across VLANs.

### Test 1 (pinging same VLAN and Different VLAN):

- Ping to 195.1.20.4 (same VLAN, VLAN 20) → Success, 4/4 replies received, 0% loss

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 195.1.20.4

Pinging 195.1.20.4 with 32 bytes of data:

Reply from 195.1.20.4: bytes=32 time=1ms TTL=128
Reply from 195.1.20.4: bytes=32 time<1ms TTL=128
Reply from 195.1.20.4: bytes=32 time<1ms TTL=128
Reply from 195.1.20.4: bytes=32 time<1ms TTL=128

Ping statistics for 195.1.20.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 195.1.10.2

Pinging 195.1.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 195.1.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

- Ping to 195.1.20.1 (different VLAN, VLAN 20) → Request timed out (100% loss)

## Test 2 (pinging same VLAN and Different VLAN)

- Ping to 195.1.10.4 (same VLAN, VLAN 10) → Success, 4/4 replies received, 0% loss
- Ping to 195.1.20.1 (different VLAN, VLAN 20) → Request timed out (100% loss)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 195.1.10.4

Pinging 195.1.10.4 with 32 bytes of data:

Reply from 195.1.10.4: bytes=32 time<1ms TTL=128
Reply from 195.1.10.4: bytes=32 time<1ms TTL=128
Reply from 195.1.10.4: bytes=32 time<1ms TTL=128
Reply from 195.1.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 195.1.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 195.1.20.1

Pinging 195.1.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 195.1.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

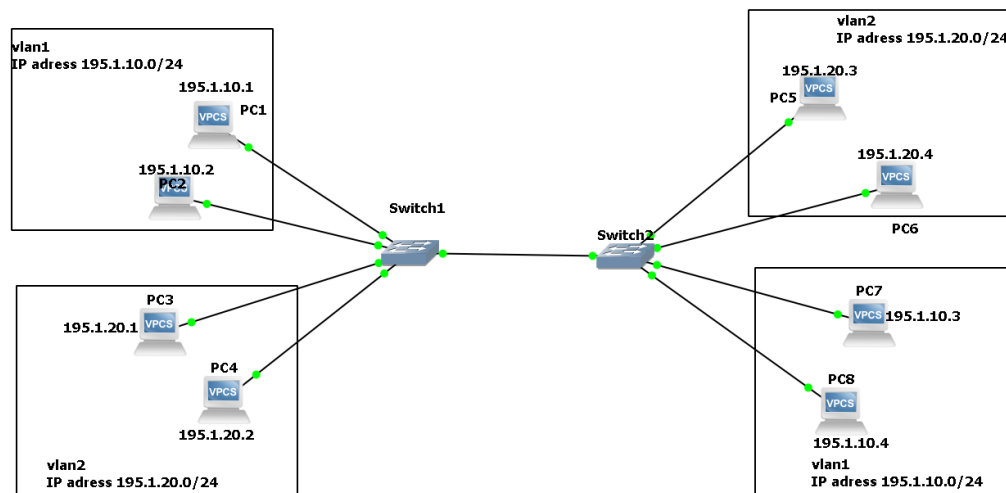
C:\>
```

## Network topology in GNS3:

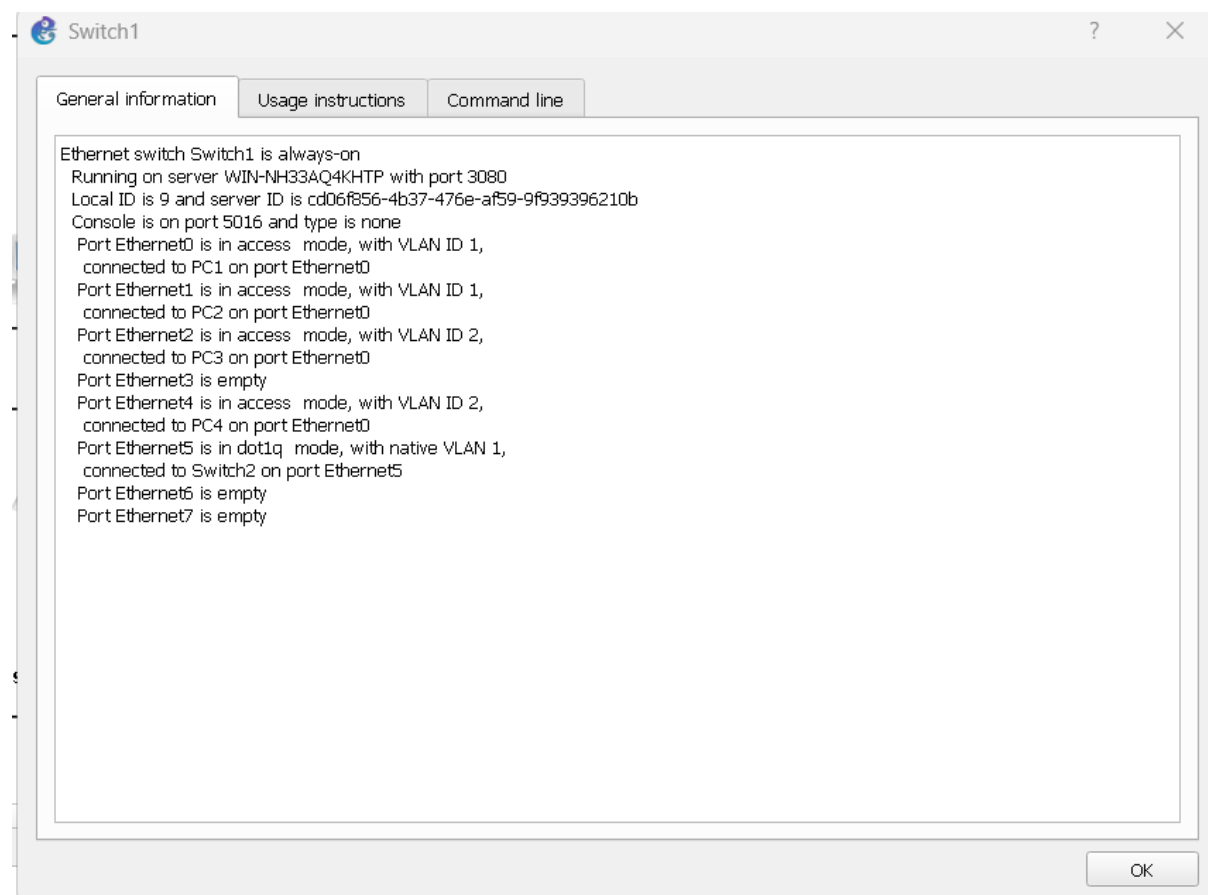
The topology consists of two switches (Switch1 and Switch2) connected via a trunk link with four PCs attached to each switch. Two VLANs were configured: VLAN 10 and



VLAN 20, each spanning both switches.

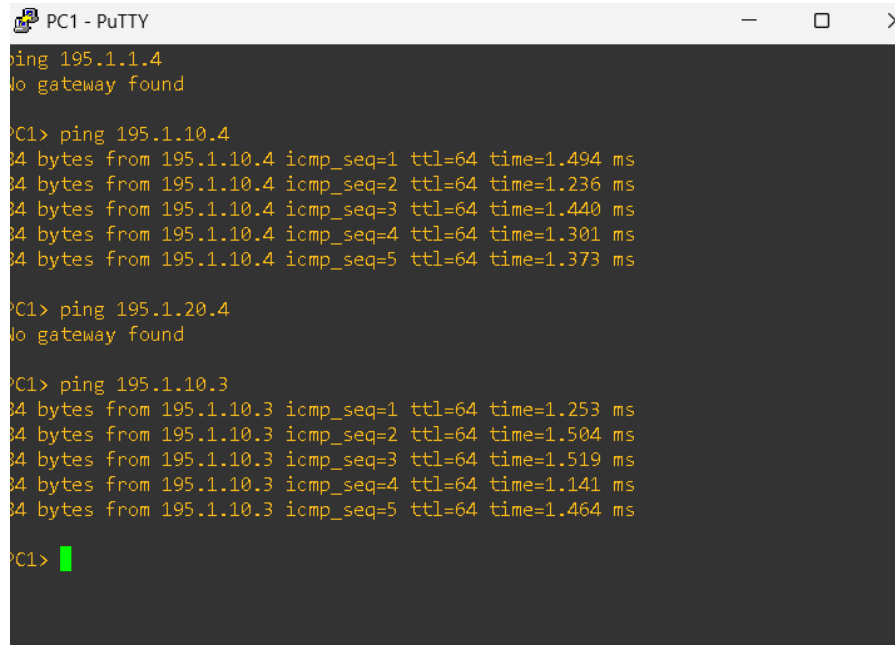


## VLANs Showed from the switch:



## Ping Tests Results:

**PC1 (VLAN 1, 195.1.10.1)** successfully pinged 195.1.10.4 (PC8) and 195.1.10.3 (PC7) — both in VLAN 1 — but got **"No gateway found"** when attempting to ping 195.1.1.4 and 195.1.20.4 (VLAN 2), since no gateway/route exists to reach a different subnet without inter-VLAN routing.



```
PC1 - PuTTY
ping 195.1.1.4
No gateway found

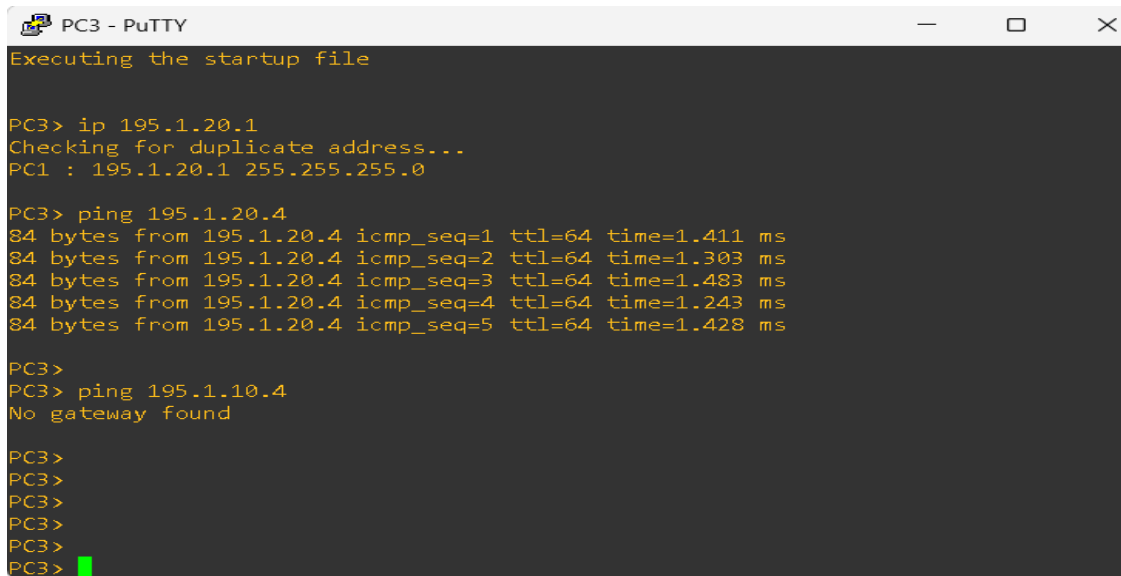
PC1> ping 195.1.10.4
84 bytes from 195.1.10.4 icmp_seq=1 ttl=64 time=1.494 ms
84 bytes from 195.1.10.4 icmp_seq=2 ttl=64 time=1.236 ms
84 bytes from 195.1.10.4 icmp_seq=3 ttl=64 time=1.440 ms
84 bytes from 195.1.10.4 icmp_seq=4 ttl=64 time=1.301 ms
84 bytes from 195.1.10.4 icmp_seq=5 ttl=64 time=1.373 ms

PC1> ping 195.1.20.4
No gateway found

PC1> ping 195.1.10.3
84 bytes from 195.1.10.3 icmp_seq=1 ttl=64 time=1.253 ms
84 bytes from 195.1.10.3 icmp_seq=2 ttl=64 time=1.504 ms
84 bytes from 195.1.10.3 icmp_seq=3 ttl=64 time=1.519 ms
84 bytes from 195.1.10.3 icmp_seq=4 ttl=64 time=1.141 ms
84 bytes from 195.1.10.3 icmp_seq=5 ttl=64 time=1.464 ms

PC1> 
```

**PC3 (VLAN 2, 195.1.20.1)** successfully pinged 195.1.20.4 (PC6), also in VLAN 2, but received **"No gateway found"** when pingging 195.1.10.4 (VLAN 1).



```
PC3 - PuTTY
Executing the startup file

PC3> ip 195.1.20.1
Checking for duplicate address...
PC1 : 195.1.20.1 255.255.255.0

PC3> ping 195.1.20.4
84 bytes from 195.1.20.4 icmp_seq=1 ttl=64 time=1.411 ms
84 bytes from 195.1.20.4 icmp_seq=2 ttl=64 time=1.303 ms
84 bytes from 195.1.20.4 icmp_seq=3 ttl=64 time=1.483 ms
84 bytes from 195.1.20.4 icmp_seq=4 ttl=64 time=1.243 ms
84 bytes from 195.1.20.4 icmp_seq=5 ttl=64 time=1.428 ms

PC3>
PC3> ping 195.1.10.4
No gateway found

PC3>
PC3>
PC3>
PC3>
PC3>
PC3> 
```

**Conclusion:**

This report has demonstrated the practical application of FLSM and VLSM subnetting across a variety of small-network scenarios, showing how bit-borrowing directly shapes subnet masks, host capacity, and address ranges. The FLSM examples demonstrated simplicity but relative inefficiency, while the VLSM examples showed how using subnet sizes that match actual host requirements minimizes address wastage. VLAN implementation and ping tests further verified that devices on the same VLAN can freely communicate and that traffic between VLANs is blocked unless a Layer 3 device is used to route between the VLANs. To summarize, the exercises highlight that a successful network design is based on efficient IP address planning as well as appropriate traffic segmentation.